

Lecture 08: Inter-process Communication in Shared Memory

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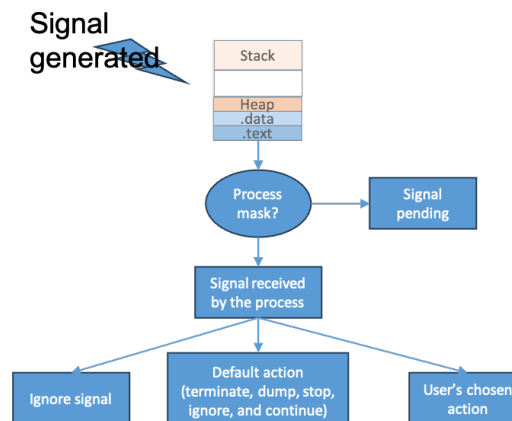
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Last Lecture

```
int global=0;
int create_process_and_run(char* command) {
    int status = fork();
    if(status < 0) {
        printf("Something bad happened\n");
    } else if(status == 0) {
        printf("I am the child process\n");
        global++;
    } else {
        printf("I am the parent Shell\n");
    }
    printf("Global value = %d\n",global);
    ....
    return 0;
}
```

```
int create_process_and_run(char* command) {
    int status = fork();
    if(status < 0) {
        printf("Something bad happened\n");
        exit(0);
    } else if(status == 0) {
        printf("I am the child process\n");
        char* args[2] = {"./fib", "40"};
        execv(args[0], args);
        printf("I should never print\n");
    } else {
        printf("I am the parent Shell\n");
    }
    ....
    return 0;
}
```



- Process creation
 - Copy on write when forked
- Child process should call **exit** system call before terminating to pass its exit status to the parent process
 - Parent can retrieve it by calling **wait**
- Zombies and orphan processes
- The exec system call

Today's Class

- Inter-process communication
 - Signals
 - Pipes
 - Shared memory

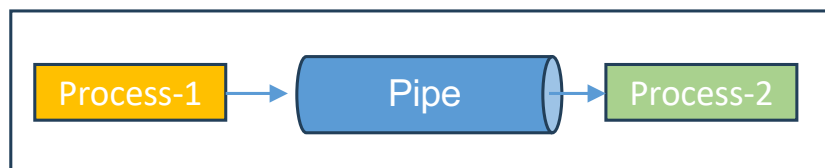
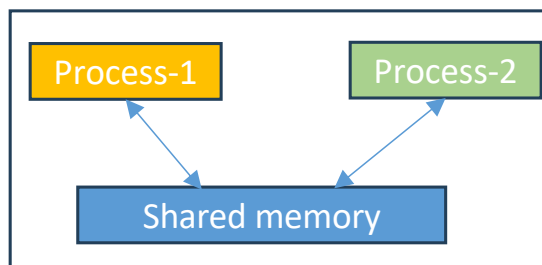
The OS Story Evolving So Far...

1. A **cook** (OS) manages a **kitchen** (*computer*)
2. The job of a cook is to do **cooking** (*execute binaries*) by reading **recipes** (*executable file*) in a **standard format** (*ELF*)
3. The cook has received recipe of a dish in **different files** (*relocatable files*). It will have to use a **stapler** (*linker*) to stitch these files to make a final recipe
4. The cook has to **store** (*load*) the **steps** (*instructions*) from the recipe in his memory to start cooking. The cook then execute instructions **one by one** (*call stack*) for cooking
5. The kitchen has gained a lot of popularity and have started receiving lot of recipes for cooking. The cook has hired several **helpers** (*processes*), where one helper cooks one recipe
6. To ensure the helpers don't fight for kitchen resources and make a mess, the ground rule is set that all helpers must talk to the head cook via a **hotline** (*interrupts*) if they need to access any kitchen resources
7. One of the helper got a huge recipe to cook. He decided to create an **identical twin** (*fork*) to let him own the **preparation** (*exec*) of one of the sub recipe

The OS Story Evolving So Far...

8. The identical twin of a helper and the helper himself must be able to **communicate** (*inter-process communication*) with each other without disturbing other helpers for properly cooking the recipe

Inter-Process Communication



- Recall, forked process calls an exec after which it has its own address space that is not shared with the parent
 - *Protection provided by OS*
- IPC mechanisms to share information between processes

Points to Ponder

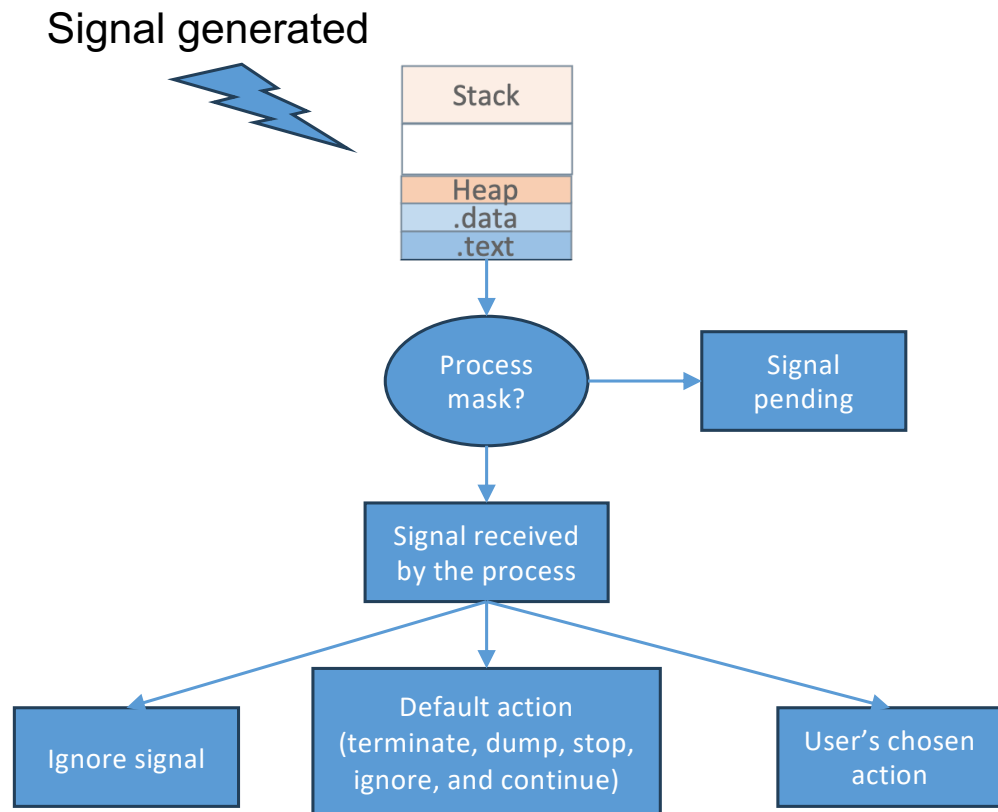
```
vivek@possum:~/os23$ vi infinite-fib.c
vivek@possum:~/os23$ gcc infinite-fib.c
vivek@possum:~/os23$ ./a.out
Input i:
20
Fib(20) = 6765
Input i:
^C
vivek@possum:~/os23$ ./a.out
Input i:
20
Fib(20) = 6765
Input i:
30
Fib(30) = 832040
Input i:
^Z
[1]+  Stopped                  ./a.out
vivek@possum:~/os23$ fg
./a.out
23
Fib(23) = 28657
Input i:
█
```

- What happens when we press the following key combination on a shell that is running some program
 - Ctrl-c
 - Process simply terminates even though it was executing some instruction inside the .text segment
 - Ctrl-z
 - Process moves to background (you can get it back to foreground using "fg")
 - How come these actions are happening even though the program did not have any code to demonstrate the above mentioned behavior
- How does a child process that died inform about its fate to its parent?
- What happens when a program attempts to access an invalid memory address?

Signals – A Limited Form of IPC

- Signals (*software interrupts*) are technique used by the OS to communicate with a process
- Every signal has a name that has a unique number assigned
 - Ctrl-c and Ctrl-z generates software interrupt that is handled by the OS by sending SIGINT (2) and SIGSTOP (20) signal, respectively, to the running process
- Each signal type has a default handler
 - A program can install its own handler for signals of any type
 - Except SIGKILL (default action is to exit the process) and SIGSTOP (default action is to suspend the process)
 - Unlike SIGKILL, SIGINT (Ctrl-c) is a polite way to terminate a process

Signal Delivery and Handling



- Process can use `sigprocmask` function to make changes to its signal mask (member in `task_struct`) if it wants to temporarily block a set of signals (**not all signals can be blocked, e.g., SIGKILL and SIGSTOP**)
- User can create his own signal handler function (we will see soon)
- Signal name has prefix “SIG”. Some of the signals that you can easily come across are:
 - SIGINT (ctrl-c), SIGQUIT (ctrl-\), SIGSTOP (ctrl-z), SIGTRAP (breakpoint), SIGSEV (segmentation fault), SIGTERM (process termination), SIGCHLD (child stopped/terminated), SIGFPE (floating point exception), etc.

Catching the Signal

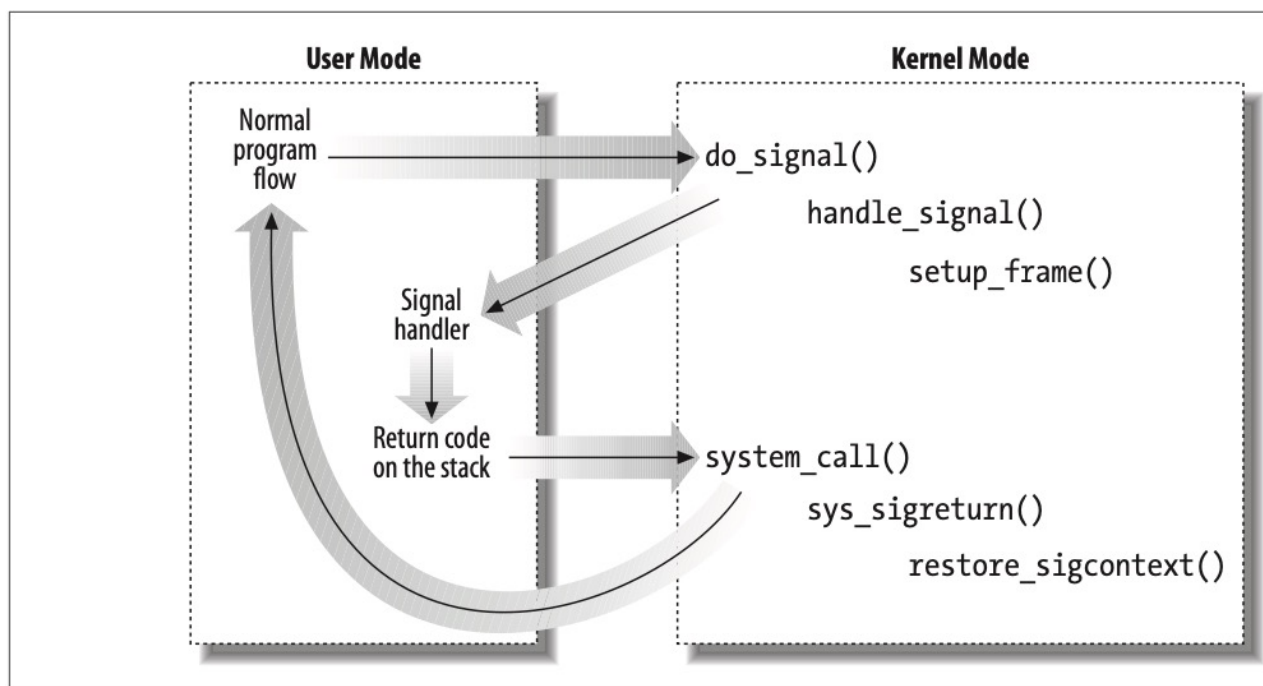


Figure 11-2. Catching a signal

- In the absence of user signal handler, signal is handled in kernel mode, otherwise it is handled in the user mode
 - See the exact sequence as shown in the Figure when user has provided the signal handler (usermode → kernelmode → usermode → kernelmode → usermode)
- Process switches into kernel mode after receiving a non-blocking signal
- Kernel handle the signal by first setting up stack frame on user stack (save context)
- Control is passed to user stack and user signal handler is executed
- Control is returned back into kernel mode which restores the user stack to its original state to resume the execution of the program
 - **sigreturn** system call used to restore process state

Programming Signals

Register handler using
sigaction syscall

Why Fib(20) is
calculated twice?

```
static void my_handler(int signum) {
    static int counter = 0;
    if(signum == SIGINT) {
        char buff1[23] = "\nCaught SIGINT signal\n";
        write(STDOUT_FILENO, buff1, 23);
        if(counter++==1) {
            char buff2[20] = "Cannot handle more\n";
            write(STDOUT_FILENO, buff2, 20);
            exit(0);
        }
    } else if (signum == SIGCHLD) {
        char buff1[23] = "Caught SIGCHLD signal\n";
        write(STDOUT_FILENO, buff1, 23);
    }
}
```

```
int main() {
    struct sigaction sig;
    memset(&sig, 0, sizeof(sig));
    sig.sa_handler = my_handler;
    sigaction(SIGINT, &sig, NULL);
    sigaction(SIGCHLD, &sig, NULL);
    int n;
    while(1) {
        printf("Input i: \n");
        scanf("%d",&n);
        if(fork()==0) {
            printf("Fib(%d) = %d\n",n,fib(n));
            exit(0);
        }
        else wait(NULL);
    }
    return 0;
}
```

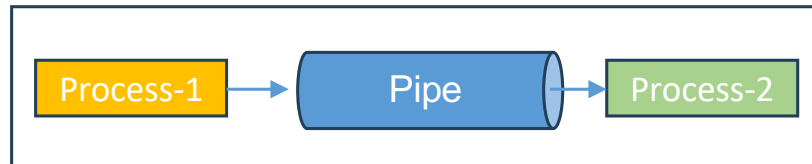
```
vivek@possum:~/os23$ ./a.out
Input i:
10
Fib(10) = 55
Caught SIGCHLD signal
Input i:
20
Fib(20) = 6765
Caught SIGCHLD signal
Input i:
^C
Caught SIGINT signal
Fib(20) = 6765
Caught SIGCHLD signal
Input i:
^C
Caught SIGINT signal
Cannot handle more
```

- **sigaction** is used to change the default action associated with a signal
 - Except SIGKILL and SIGSTOP
- User can assign his own signal handler method that would get invoked when the assigned signal is received by the process
- Only asynchronous safe functions should be called inside signal handler
<https://man7.org/linux/man-pages/man7/signal-safety.7.html>

Today's Class

- Inter-process communication
 - Signals
 - Pipes
 - Shared memory

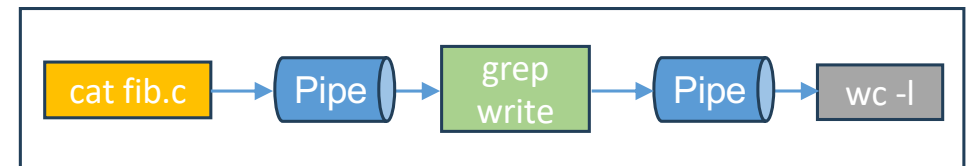
IPC Using Pipes



- Pipes (analogous to water pipe) is a unidirectional stream of data flowing from a source process to a destination process
 - Kernel buffer exposed to processes as a pair of file descriptors (readable end and writable end). Default buffer size is 16 pages (16x4096 bytes)
 - Data delivered in the same order as sent
 - Uses blocking IO and the writer process will block if the pipe is full
- We use it frequently on Shell
 - Better than using temporary files as pipes automatically clean up themselves unlike files that must be explicitly removed using the “rm” command on Shell

```

vivek@possum:~/os23$ cat fib.c | wc -c
1788
vivek@possum:~/os23$ cat fib.c | wc -l
77
vivek@possum:~/os23$ cat fib.c | grep write | wc -l
10
  
```



Programming With pipe

```
int main() {
    int fd[2], status;
    pipe(fd);
    if(fork() == 0) {
        /* Child process */
        close(fd[0]);
        char buff[] = "Hello my dear good Parent";
        sleep(2);
        write(fd[1], buff, sizeof(buff));
        exit(0);
    }
    /* Parent process */
    close(fd[1]);
    char buff[100];
    read(fd[0], buff, sizeof(buff));
    printf("My obedient child says: %s\n", buff);
    wait(NULL);
    return 0;
}
```

- **pipe** expects an int array of size two only for a readable and writable file descriptor
- Reader and writer processes must close the pipe end they are not going to use
 - It is very important as fork duplicates the open file descriptors in the parent process
 - Unclosed file descriptors might lead to incomplete or incorrect data transfer
- What would happen if the child sleeps before writing to the pipe, as the parent has already issued the read?
- **dup2** (code not shown here)
 - Used for duplicating a file descriptor
 - E.g., duplicate one of the end of the pipe as STDOUT or STDIN
 - Duplicating to STDOUT will cause printf to print to the pipe instead of the STDOUT
 - Used by the Shell when we pipe the output of one command to another command

Today's Class

- Inter-process communication
 - Signals
 - Pipes
 - Shared memory

Array Sum Program

```
int main() {  
    int A[SIZE]; //initialized  
    int sum=0;  
    for(int i=0; i<SIZE; i++) sum += A[i];  
    return 0;  
}
```

- A simple program to calculate the sum of all the elements inside a huge array
- How to shorten the execution time for calculating the sum for a very big array?

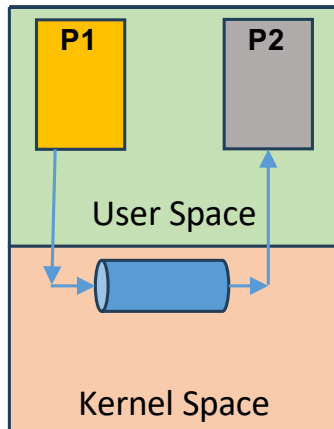
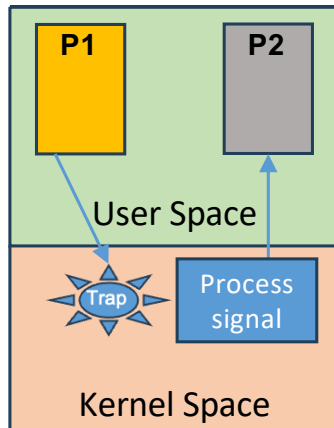
Array Sum Program

```
int main() {
    int A[SIZE]; //initialized
    int sum1=0, sum2=0;
    if(fork() == 0) {
        for(int i=SIZE/2; i<SIZE; i++) sum1 += A[i];
        exit(0);
    }
    for(int i=0; i<SIZE/2; i++) sum2 += A[i];
    wait(NULL);
    return 0;
}
```

Can we do $\text{sum1} + \text{sum2}$ in any of the two process to get the total sum?

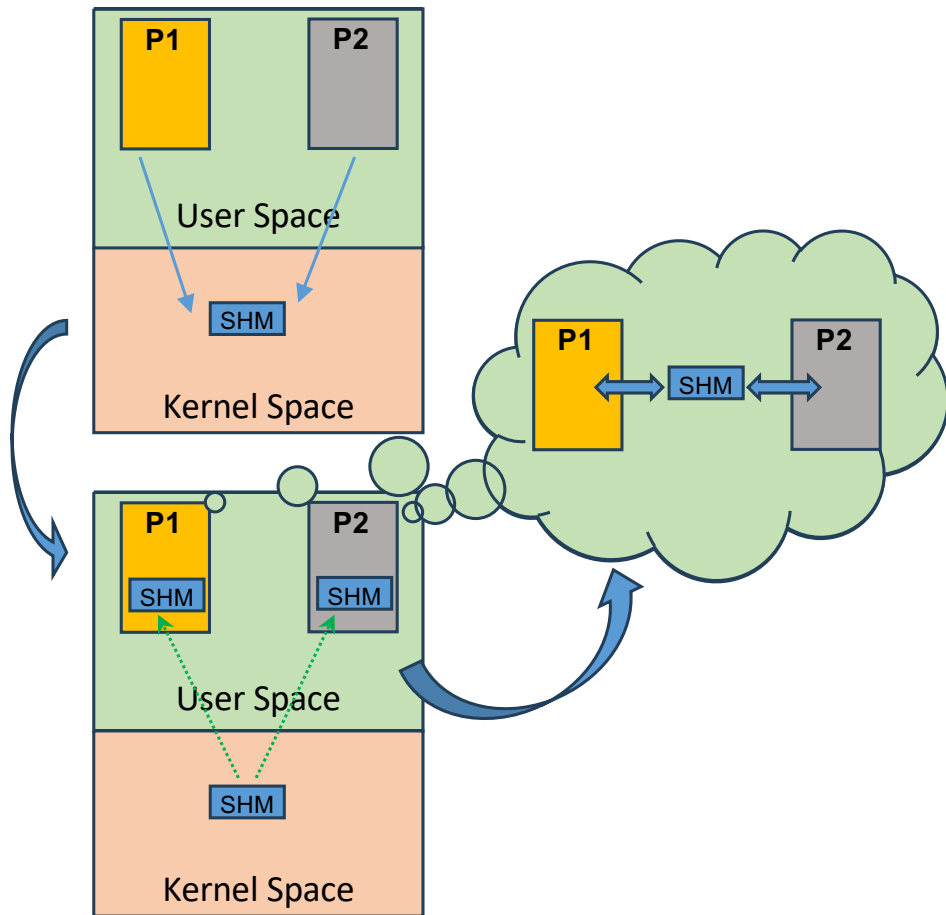
- We can fork a child and let the parent and child calculate the sum of left and right half of the array, respectively
- **The problem(s)**
 - How the child and parent would communicate their partial results to get the total sum?
 - Can the child pass the partial sum as its exit code?
 - `exit` API accepts only "int" type parameter. We cannot pass the sum of "double" type array, etc.
 - Can we setup a pipe between the parent and child?
 - Yes, but what if there are several children to improve the execution time further? We will have to use multiple pipes and keep track of them (closing fds differently at child v/s parent, etc.)
 - What if its some other kind of program, such as vector addition or matrix multiplication?
 - Three arrays (or matrices) would require accesses by child and parent

IPC Using Signals and Pipes



- Issues with signaling
 - Limited form of IPC
 - Requires help from OS for every signaling instances
 - Transition between user space and kernel space (overheads)
- Issues with pipes
 - Limited buffer size (although significantly better than signaling)
 - Use of system calls for every communication
 - Transition between user space and kernel space (overheads)
 - One way communication
 - Blocking communication

IPC Using POSIX Shared Memory



- Shared memory is the fastest form of IPC in a single machine in which processes can asynchronously write to a shared region of memory **without** the need for switching to kernel space
 - User specified memory size
- Access to the shared memory is achieved as follows
 1. Create and open a new shared memory object using `shm_open` system call in the parent process
 2. Set the desired size for the newly created shared memory region using `ftruncate`
 3. Map the shared memory region into the process's address space using `mmap` (parent process calls `mmap` that gets shared in child's address space after `fork`)
 4. Both child and the parent processes un-maps the shared memory from their address space upon completion by using `munmap` system call, followed by closing the file descriptor for the shared memory object (`close` system call)
 5. Parent deletes the shared memory object by using `shm_unlink`
- Can be used for parallel programming!

Array Sum Program

```
int main() {
```

```
typedef struct shm_t {  
    int A[SIZE];  
    int sum1;  
    int sum2;  
} shm_t;
```

- We will create a shared memory segment of our own type `shm_t` that contains the array and a variable `sum` to store the partial result
 - Although, its not necessary to include the array `A` inside `shm`
 - Why?

```
    return 0;
```

```
}
```

Array Sum Program

```
int main() {
    shm_t* shm = setup();
```

1) shm_open (...)
2) ftruncate (...)
3) shm = mmap (...)

```
typedef struct shm_t {
    int A[SIZE];
    int sum1;
    int sum2;
} shm_t;
```

```
cleanup();
return 0;
```

1) munmap (shm, ...)
2) close (...)
3) shm_unlink (...)

```
}
```

- The three steps for setting up a shared memory segment are:
 - Creating a shared memory object (shm_open)
 - Assigning a size for the shared memory object (ftruncate)
 - Mapping the shared memory into the process's address space (mmap)
- The three steps in cleanup are:
 - Removing the mapping of shared memory from process's address space
 - Closing the file descriptor for the shared memory object
 - Deleting the shared memory object

Array Sum Program

```
int main() {
    shm_t* shm = setup();
    if(fork()==0) {
```

```
typedef struct shm_t {
    int A[SIZE];
    int sum1;
    int sum2;
} shm_t;
```

```
        cleanup_and_exit();
    } else {
```

- 1) munmap (shm, ...)
- 2) close (...)
- 3) exit(0)

```
        wait(NULL);
```

```
    }
    . . . . .
    cleanup();
    return 0;
}
```

- Parent process will fork children that gets a copy of the parent's address space (inherits the file descriptor and shared memory object from the parent)
 - Reason for not including array A in shared memory
- Child process also carries out cleanup similar to parent except for calling **shm_unlink** that is called only by the parent
- If parent process fail to call **shm_unlink**, then the **shm_open** will fail the next time (unless you change the name of the memory region)
- Child must also call **exit** in their cleanup call
 - Why?

Array Sum Program (Version 1)

```
int main() {
    shm_t* shm = setup();
    if(fork()==0) {
        int local=0;
        for(int i=0; i<SIZE/2; i++) shm->sum1+=shm->array[i];
        cleanup_and_exit();
    } else {
        int local=0;
        for(int i=SIZE/2; i<SIZE; i++) shm->sum2+=shm->array[i];
        wait(NULL);
    }
    int total = shm->sum1 + shm->sum2;
    cleanup();
    return 0;
}
```

```
typedef struct shm_t {
    int A[SIZE];
    int sum1;
    int sum2;
} shm_t;
```

- Both child and the parent process calculate the partial sum independently
- Parent process combine the partial sum from the child to its own calculation to get the total
 - Done only after **wait**

Next Lecture

- Semaphores